A Comparative Study of Neural Networks and Physics-Informed Neural Networks for Stellar Property Modelling

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April 2026

Abstract

Modelling physical systems from observational data is a fundamental problem in astrophysics, where analytical solutions are often unavailable and data is frequently incomplete or noisy. While neural networks (NNs) provide a powerful data-driven approach for approximating complex relationships, their reliance on data alone can limit their ability to generalise in regions where data is sparse or absent. Physics-informed neural networks (PINNs) address this limitation by incorporating prior knowledge of the underlying physical behaviour into the training process. In this study, a systematic comparison is performed between a standard NN and a PINN for modelling stellar properties using simulated stellar evolution datasets. Both models are trained under equivalent conditions and evaluated across three data regimes: within the training domain, interpolation across sparsely sampled regions, and extrapolation beyond the range of the training data. Model performance is assessed using statistical error metrics, including RMSE, MAE, and R^2, alongside smoothness-based measures and visual diagnostics to evaluate physical consistency. The results show that while the standard NN achieves slightly lower prediction error within the training domain, its performance degrades significantly in interpolation and extrapolation, where it produces simplified, physically invalid linear predictions. In contrast, the PINN maintains lower prediction error in these regimes and produces smoother, more physically consistent outputs. These findings demonstrate that incorporating physics-informed constraints improves generalisation, highlighting their importance for reliable scientific modelling in data-limited astrophysical applications.

1 Introduction

In physics, observational data is used to infer the properties and behaviour of physical systems, from laboratory-scale experiments to large-scale astrophysical phenomena. Traditionally, this is achieved by modelling the data using equations derived from first principles that describe the underlying physics, followed by statistical methods—often within a Bayesian framework [1]—to estimate model parameters and their associated uncertainties. This approach has been highly successful when the governing equations are well understood and computationally tractable. In practice, many systems are computationally expensive to simulate, and analytical solutions are often not available outside of idealised cases. In addition, particularly in astrophysical settings, observational data is often noisy, incomplete, and non-repeatable [2]. Despite these challenges, it remains essential to reliably infer physical parameters from such data. This motivates the exploration of modelling approaches that can operate effectively under imperfect data conditions while remaining consistent with known physical principles.

As an alternative, neural networks (NNs) are powerful data-driven tools for modelling complex systems [3]. They are mathematical models that are able to learn and approximate functions from data, even when the true underlying behaviour is unknown [4]. They are highly effective at learning nonlinear relationships [5] and have demonstrated strong performance on regression tasks [6]. Furthermore, NNs have proven effective in modelling datasets subject to significant uncertainty and noise, particularly in astrophysical settings [7]. However, the data-driven nature of standard NNs poses its own limitations. Due to their dependence on data, it has been shown that they perform well where data is readily available but often struggle when required to interpolate across sparsely sampled regions [8] or extrapolate beyond the data [9]. NNs are merely mathematical models and have no knowledge of the underlying physical laws that govern the data. In a physics context, where observations are frequently incomplete or sparse and where reliable extrapolation and interpolation is often essential, relying solely on data without incorporating physical understanding can therefore lead to significant errors and instability.

Physics-informed neural networks (PINNs) address these challenges by not only relying solely on the data but also incorporating knowledge of the underlying physical behaviour of the system [10]. This approach embeds scientific knowledge into its predictions, allowing the model to remain consistent with the underlying physics even when poor data conditions, such as noise or missing regions, may lead it astray. As a result, PINNs can often outperform standard NNs in tasks involving extrapolation or interpolation across missing datasets, as well as in datasets involving high levels of uncertainty, giving PINNs a higher degree of robustness and a superior generalisation ability [11] compared to their standard counterparts. This makes PINNs particularly appealing for physics applications, where observational data is often limited and noisy. However, the added complexity of PINNs introduces several challenges. Recent research has found that PINNs are highly sensitive to their internal parameters and hence face difficulties in optimisation [12]. In addition, due to their incorporation of physics, they have been found to be more computationally intensive [13], resulting in slower prediction times. One approach that has been proposed to address these optimisation difficulties is the use of adaptive loss-weighting methods, which dynamically adjust the balance between the data-fitting and physics-based components of the training objective [14].

Despite the rapid development of PINNs, several important questions remain unresolved.

In particular, relatively limited work provides direct and systematic comparisons between PINNs and standard NNs across different data regimes, such as sparse sampling, interpolation across missing regions, and extrapolation beyond the training domain. A large portion of the literature is devoted to the application of PINNs in modelling specific physical systems [15]. As a result, it is still not fully clear under what conditions physics-informed training provides a consistent advantage over purely data-driven NNs, particularly in realistic observational settings where data is noisy and incomplete.

This study seeks to address these open questions through a systematic comparison of PINNs and standard NNs using simulated astrophysical datasets derived from complex stellar evolution models. Both models are trained under equivalent conditions to enable a controlled evaluation of their performance. The comparison is carried out across different data regimes, including regions of high data density, interpolation across missing regions, and extrapolation beyond the training domain, in order to assess their generalisation abilities. This evaluation is repeated across multiple datasets to ensure that the results are consistent and not specific to a single data representation. Model performance is assessed in terms of both predictive accuracy and physical consistency. Through this analysis, the study aims to determine the conditions under which PINNs provide a meaningful advantage over standard NNs in astrophysical applications.

2 Theoretical Background

2.1 Standard Neural Networks

A NN learns a mapping between inputs and outputs that approximates the behaviour present in the data it is trained on. It is composed of an input layer, one or more hidden layers, and an output layer, where each layer contains a set of nodes, with the number of nodes in the input and output layers corresponding to the dimensionality of the inputs and outputs respectively. Each node in a layer carries a value that is transformed and combined to form the values of the nodes in the next layer, allowing the network to map inputs to a final predicted output. The values of the nodes are transformed using weights and biases, which form the model's set of internal parameters. These parameters determine the predictions produced by the network and are optimised during training so that the predictions reflect the relationships present in the data. Between each layer, a nonlinear activation function is applied. The introduction of nonlinearity is essential, as without it the network would be limited to learning only linear relationships, whereas its inclusion allows the model to capture more complex patterns in the data. The model is trained to optimise its internal parameters to minimise the difference between its predictions and the true values in the training data. This is achieved in the following way. Initially, the model is untrained and produces inaccurate predictions. A loss function is defined as the mean squared difference between the network's predicted output and the true values. Through backpropagation, the gradient of the loss with respect to all model parameters is computed by propagating error information backwards through the network. These gradients are then used by an optimiser to update the parameters according to a defined update rule:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \, \nabla_\theta \mathcal{L}(\theta) \tag{1}$$

where θ denotes the model parameters, η the learning rate, and $\mathcal{L}(\theta)$ the loss function.

Each parameter is adjusted in the direction that reduces the loss, with the learning rate controlling the size of each update. During training, this process is applied repeatedly over many epochs. This progressively reduces the loss and improves the network's performance, allowing it to learn and model the underlying relationships in the data. A NN can therefore be viewed as a collection of mathematical operations whose parameters are systematically optimised to produce the desired outputs, emulating a form of artificial learning.

2.2 Physics-Informed Neural Networks

A PINN extends the standard NN framework by incorporating physical constraints into the training process. Both models use an identical layered architecture, with the key distinction lying in how they are trained. In particular, the loss function in a PINN is not defined as solely the loss from the data, but also includes a physics loss term defined as the residual of the system's governing differential equations averaged over all training points.

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \, \mathcal{L}_{\text{physics}} \tag{2}$$

where $\mathcal{L}_{\text{data}}$ denotes the data-fitting loss, $\mathcal{L}_{\text{physics}}$ the physics-based residual loss, and λ a weighting parameter controlling the relative contribution of the two terms.

This term acts as a loss from the system's governing differential equations being satisfied, and to evaluate it requires the PINN to compute the derivatives of its predictions with respect to the input variables, and in practice smooth activation functions are used to ensure these derivatives can be computed reliably. Minimising this loss encourages the PINN to produce predictions that align with the observed data while also satisfying the governing physical constraints, resulting in physically valid solutions. The PINN is then trained in a similar manner to a standard NN, where gradients of the loss are computed through backpropagation and the model parameters are updated using an optimiser over many epochs.

In practice, PINNs can be challenging to train. The optimisation process requires balancing multiple competing loss terms, including the data loss and the physics residual, and training can be highly sensitive to the relative weighting between these components. If this balance is not chosen appropriately, the network may either prioritise fitting the data while violating the physical constraints, or strongly enforce the physics term while failing to reproduce the observed data. In addition, PINNs can be computationally expensive due to the need to compute higher-order derivatives of the model outputs during training. As a result, training PINNs is often more complex and less stable than training standard NNs.

3 Methods

This study presents a systematic comparison between a standard NN and a PINN for modelling stellar properties across multiple simulated stellar evolution datasets. For each dataset, both models are trained to learn mappings between selected stellar input parameters and corresponding output properties. Both models are trained under equivalent

conditions to allow for direct comparison. Performance is assessed in terms of predictive accuracy and physical consistency using a combination of quantitative metrics and visual analysis. In addition, performance is evaluated across different data regimes: within the training domain, interpolating across missing regions, and extrapolating beyond the range of the training data. This is done to assess the models' generalisation ability, which is essential in astrophysical applications where observations are often incomplete, noisy, and sparsely sampled.

3.1 Training Data

Two simulated datasets are used to evaluate model performance. The first dataset is from a published paper [16], and consists of simulated stellar evolution tracks generated using complex stellar structure models that describe how stars evolve over time given specific initial conditions. Each track corresponds to a single star with fixed initial parameters and records its properties at multiple stages throughout its evolution. The dataset includes a wide range of stellar parameters, including global surface properties such as mass, radius, luminosity, and effective temperature, initial composition parameters such as hydrogen, helium, and metallicity, and internal properties such as central hydrogen content, density, and core energy generation rates. By systematically varying initial parameters—including stellar mass, composition, metallicity, and the mixing-length parameter—the authors constructed a grid of evolutionary tracks spanning a wide range of stellar types. Each point along a track represents a different evolutionary stage of the same star; however, for machine learning purposes, these points are treated as independent samples. This results in a large set of physically consistent training examples covering a broad region of the stellar parameter space, including both well-sampled and sparsely sampled regions.

The second dataset is obtained from the MESA Isochrones and Stellar Tracks (MIST) database [17], developed by the Harvard–Smithsonian Center for Astrophysics, and consists of simulated stellar evolution isochrones generated using the Modules for Experiments in Stellar Astrophysics (MESA) framework. These models numerically solve complex stellar structure and evolution equations to produce physically consistent predictions of stellar properties. In contrast to stellar evolution tracks, which follow a single star over time, isochrones represent populations of stars with different initial masses evaluated at a fixed age. The dataset includes parameters such as equivalent evolutionary point (EEP) (a parameterisation of stellar evolutionary stage), initial mass, age, luminosity, effective temperature, radius, surface gravity, mass loss rate, surface abundances, and internal properties such as core mass and central temperature. As with the first dataset, each point is treated as an independent sample for machine learning. The dataset spans a broad region of the stellar parameter space across a wide range of ages, masses, and compositions.

Due to the size and complexity of the datasets, a series of pre-processing steps were applied, including parameter selection and standardisation. For the first dataset, the inputs were limited to stellar mass, age, initial hydrogen fraction, initial helium fraction, metallicity, and the mixing-length parameter, with outputs defined as stellar luminosity and effective temperature. For the second dataset, the inputs were reduced to EEP, initial stellar mass, and metallicity, while the outputs included isochrone age, effective temperature, luminosity, surface gravity, and stellar radius. These selections ensure that the models focus on the most relevant parameters while maintaining computational efficiency. All

input features were standardised to have zero mean and unit variance to ensure balanced contribution during training. In addition, logarithmic transformations were applied to the variables to improve computational efficiency and numerical stability, while also ensuring physically meaningful predictions by preventing non-physical negative values for strictly positive quantities.

3.2 Neural Network Architectures

Both the standard NN and PINN were designed to learn the mappings and model the relationship between selected stellar input parameters and corresponding output properties defined for each dataset. The standard model serves as a purely data-driven baseline, while the PINN incorporates additional constraints during training to encourage physically consistent predictions. This enables a direct comparison between data-driven and physics-informed approaches.

Both networks share an identical architecture so that any differences in performance can be attributed to the inclusion of physics-informed constraints rather than differences in the network architecture. The networks contain 4 hidden layers consisting of 128, 128, 64, and 64 neurons respectively, with the hyperbolic tangent (tanh) activation function applied between layers to capture nonlinear relationships while remaining smooth and differentiable, as required by the PINN. This architecture provides sufficient representational capacity to model the nonlinear structure of the stellar parameter space while maintaining stable and efficient training. Increasing model complexity beyond this configuration offers limited improvement in predictive performance and increases computational cost and the risk of overfitting.

While PINNs are typically formulated by enforcing the governing differential equations of the system through a residual-based loss, this approach is not adopted here due to the complexity of the stellar structure equations, the lack of a suitable closed-form representation for direct incorporation into the loss function, and the increased computational cost associated with enforcing such constraints during training. Instead, a simplified physics-based constraint is introduced based on the assumption that stellar properties vary smoothly with respect to their governing parameters, which is a valid approximation for stable stellar evolution. To enforce this behaviour, a curvature-based penalty is applied using second-order derivatives of the model outputs with respect to the inputs. A threshold penalty is then applied so that only excessive curvature contributes to the loss, allowing physically meaningful variation while suppressing non-physical oscillations. The resulting physics loss is computed as a sum over all input–output combinations and is scaled by a weighting coefficient that controls the strength of the penalty. By combining the standard data-driven loss with this physics-inspired term, the PINN leverages both the information in the training data and prior knowledge about expected behaviour, improving generalisation, prediction accuracy, and physical consistency compared to a purely data-driven network.

$$\mathcal{L}_{\text{physics}} = \lambda \sum_{k=1}^{n_{\text{out}}} \sum_{j=1}^{n_{\text{in}}} \left(\frac{\partial^2 \hat{y}_k}{\partial x_j^2} \right)^2 \tag{3}$$

where $\mathcal{L}_{\text{physics}}$ is the physics loss, λ is a weighting coefficient controlling the strength of the penalty, $\hat{y}_k$ is the kth predicted output, x_j is the jth input variable, n_{out} is the number

of outputs, and n_{in} is the number of inputs.

3.3 Training Procedure

Model parameters were optimised using the Adam optimiser with a learning rate of 1×10^{-3}, and the models were trained for 10,000 epochs. To evaluate the models' ability to generalise both within and beyond the training domain, the datasets were partitioned with respect to stellar mass to create controlled interpolation and extrapolation regimes. For interpolation, 10% of the data corresponding to the central region of the stellar mass distribution was withheld from training, while for extrapolation, 5% of the data from each extreme of the stellar mass range was excluded. The remaining 80% of the data was used for training. This strategy enables the models to be evaluated on their ability to reproduce behaviour within the training domain, interpolate within sparsely sampled regions of the parameter space, and extrapolate beyond the range of the training data.

Several optimisation strategies were employed to ensure efficient and stable training. Firstly, training was performed in batches, with a batch size of 512. The training data was divided into batches and randomly shuffled at the start of each epoch, where each batch is processed sequentially by the model. Compared to full-batch training, this approach reduces memory requirements and improves computational efficiency, while the nature of batch sampling introduces variability in the optimisation process, which can improve convergence and help prevent the optimisation from becoming trapped in poor local minima. In addition, for the physics-informed model, a scheduled weighting scheme was applied to the physics loss to improve training stability and efficiency. This was implemented using an initial warm-up period in which the contribution of the physics loss was set to zero, allowing the model to first learn a rough approximation from the data. Following this, the weight of the physics term was gradually increased over training epochs, progressively guiding the model towards physically consistent solutions without destabilising the optimisation process. Such approaches are consistent with adaptive loss-weighting strategies proposed in the literature, which dynamically balance the contributions of data-driven and physics-based terms during training [14]. Furthermore, model checkpointing was used during training, where the model parameters corresponding to the lowest loss were saved and used for final evaluation, rather than the parameters from the final training epoch. This ensures that evaluation is performed using the best-performing model and reduces the impact of overfitting during later stages of training.

3.4 Evaluation Metrics

Model performance is evaluated based on both predictive accuracy and physical validity of their predictions after training. Both quantitative and qualitative analyses are used to assess and compare the performance of the standard NN and the PINN. Quantitative evaluation is carried out using statistical error metrics including the root mean squared error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2), which quantify predictive accuracy alongside smoothness-based statistics, which are used to assess physical validity. Qualitative assessment is performed using visual aids to support and validate model performance. In particular, predicted-versus-true plots and residual plots are used to assess predictive accuracy, while Hertzsprung–Russell (HR) diagrams and output parameter sweeps with respect to individual inputs are used to evaluate physical

validity and prediction smoothness. All analyses are repeated across the training, interpolation, and extrapolation regimes for each dataset to enable a consistent assessment of model generalisation.

3.4.1 Quantitative Analysis

Quantitative performance is assessed using statistical error metrics that compare model predictions with the corresponding true values. Prediction accuracy is evaluated using the RMSE and MAE, which provide complementary measures of prediction error. The RMSE is defined as the square root of the average squared error and penalises larger deviations more strongly due to its squared nature, making it more sensitive to outliers. In contrast, the MAE measures the average absolute deviation and provides a more robust indication of typical prediction error. To allow comparison across outputs with different scales, percentage error metrics are also computed relative to the mean of the true values, providing a scale-independent measure of performance. Metrics are computed for each output variable and averaged across all samples, with final values obtained by averaging across all outputs and both datasets to provide a single overall measure of model performance. In addition, R^2 is used to evaluate how well the model captures the underlying structure of the data. R^2 measures the variance of the prediction errors relative to the natural variance of the data, indicating the extent to which the model captures the underlying structure rather than simply approximating the mean. Values of R^2 close to 1 indicate that the model captures the dominant structure in the data, whereas values close to 0 indicate that the model fails to explain the variance and performs no better than predicting the mean.

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{4}$$

where y_i denotes the true values, $\hat{y}_i$ the predicted values, $\bar{y}$ the mean of the true values, and N the number of samples.

To assess the physical consistency of the model predictions, physically realistic outputs are expected to vary smoothly with respect to the inputs. To quantify this, two metrics are computed based on differences between adjacent prediction values along a one-dimensional input sweep, in which one input is varied while the others are held fixed. The first metric is the first difference, defined as the change in output between neighbouring points, which measures the overall variation in the prediction. The second metric is the second difference, defined as the change in these first differences, and therefore captures the curvature of the prediction. These metrics are computed for each predicted output and input sweep and are averaged over all evaluated points to provide a consistent measure of smoothness across the model predictions. Smoother predictions correspond to smaller values of the second difference.

3.4.2 Visual Analysis

Visual diagnostics are used to complement quantitative metrics by providing insight into the structure of prediction errors and the physical behaviour of the model outputs. They enable the identification of variations in performance across the parameter space and deviations from physically consistent behaviour.

Several plots are used to assess predictive performance. Predicted-versus-true plots are used to evaluate overall agreement between model predictions and the corresponding true values, where accurate predictions lie close to the one-to-one line. Residual plots, defined as the difference between predicted and true values, are plotted against the true values to identify systematic errors and regions where model performance degrades. Training loss curves, showing loss as a function of training epochs, are examined to assess the optimisation process and ensure stable convergence during training.

Moreover, model predictions are examined using plots to assess for physical validity. In particular, HR diagrams are constructed to determine whether the models reproduce the expected stellar evolution structure, providing a direct check on the physical realism of the predictions. In addition, plots of model outputs with respect to individual input variables are generated, where one input is varied while the others are held fixed, to examine how predictions change across the input space. These plots are used to assess the smoothness of the predicted relationships and to identify any non-physical behaviour, such as irregular trends or oscillations.

4 Results

The performance of the standard NN and PINN is evaluated using quantitative error metrics, smoothness measures, and qualitative visual diagnostics. Results are presented across the training domain, interpolation within sparsely sampled regions, and extrapolation beyond the training range, allowing comparison of predictive accuracy, generalisation, and physical consistency under different data conditions.

4.1 Quantitative Results

Predictive accuracy is evaluated using the RMSE, MAE, and R^2, alongside smoothness measures based on first- and second-order differences of the model predictions to assess physical consistency. As the output variables span different numerical scales, percentage error metrics are also computed relative to the mean value of each parameter to enable consistent comparison across outputs. These metrics are calculated for each output variable and averaged to provide an overall measure of performance for both models, and are further averaged across both datasets to obtain a single set of results for comparison. The prediction accuracy results are summarised in Table 1, with smoothness measures presented separately in Table 2.

Regime	Model	RMSE (%)	MAE (%)	R^2
Within Domain	NN	0.10	0.06	0.9999
	PINN	0.12	0.07	0.9998
Interpolation	NN	0.36	0.23	0.992
	PINN	0.25	0.16	0.996
Extrapolation	NN	0.82	0.51	0.958
	PINN	0.34	0.21	0.986

Table 1: Comparison of NN and PINN predictive performance across different data regimes.

The results presented in Table 1 show clear differences in predictive performance between the standard NN and the PINN across the three evaluation regimes. Within the training domain, the standard NN achieves the lowest prediction error, with RMSE and MAE values of 0.10% and 0.06%, compared to 0.12% and 0.07% for the PINN. Both models achieve R^2 values close to 1, indicating strong agreement with the underlying data where training samples are available.

In the interpolation regime, the performance of the standard NN degrades, with errors increasing to 0.36% and 0.23%, while the PINN maintains lower values of 0.25% and 0.16%. This trend becomes more pronounced in the extrapolation regime, where the standard NN exhibits a substantial increase in error, reaching 0.82% RMSE and 0.51% MAE. In contrast, the PINN retains significantly lower errors of 0.34% and 0.21%, alongside higher R^2 values. Overall, these results show that while the NN performs best within the training domain, the PINN provides improved accuracy in regions where data is limited or absent.

Model	First Difference	Second Difference
NN	0.13	0.029
PINN	0.11	0.015

Table 2: Comparison of smoothness metrics for NN and PINN predictions based on first- and second-order differences.

The smoothness metrics presented in Table 2 further highlight differences in model behaviour. As the absolute magnitude of these values is not directly meaningful, their relative comparison is used to assess the smoothness of the model outputs. The PINN exhibits lower second-difference values than the standard NN, 0.015 compared to 0.029, indicating reduced curvature and smoother variation in the predicted relationships. A similar trend is observed for the first-difference metric, where the PINN achieves slightly lower values, 0.11 compared to 0.13, although the difference is less pronounced. Overall, these results indicate that while both models capture the general trends in the data, the PINN produces smoother predictions with fewer small-scale variations.

4.2 Visual Analysis

Visual analysis is used to complement the quantitative results by providing insight into the structure of prediction errors and the physical behaviour of the model outputs for both the standard NN and the PINN. Predicted-versus-true plots are presented to assess predictive accuracy, alongside HR diagrams and one-dimensional input sweeps to evaluate physical consistency. The results are examined across the training, interpolation, and extrapolation regimes to evaluate both predictive accuracy and physical validity.

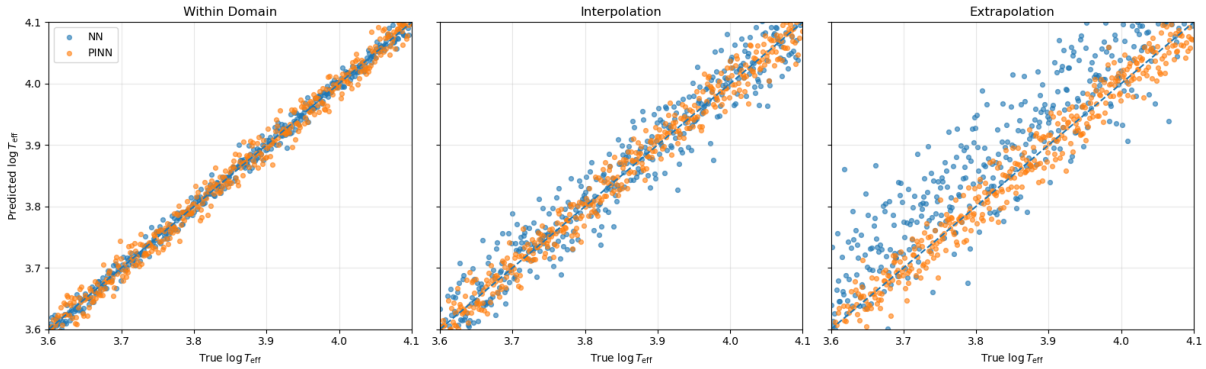


Figure 1: Predicted versus true $\log T_{\rm eff}$ for the NN and PINN across the within-domain, interpolation, and extrapolation regimes for the simulated stellar tracks dataset. A representative subset of points is shown for clarity.

Figure 1 shows the predicted-versus-true values for the effective temperature for both the standard NN and the PINN for the different data regimes. Predictions that lie closer to the one-to-one line indicate higher predictive accuracy. Within the training domain, the standard NN exhibits slightly tighter clustering, indicating a closer fit to the data, although both models perform similarly in this regime. In contrast, in the interpolation and extrapolation regions, the PINN shows more tightly clustered predictions, indicating improved predictive accuracy outside the training domain. The difference between the models is clear in interpolation and becomes more pronounced in the extrapolation regime.

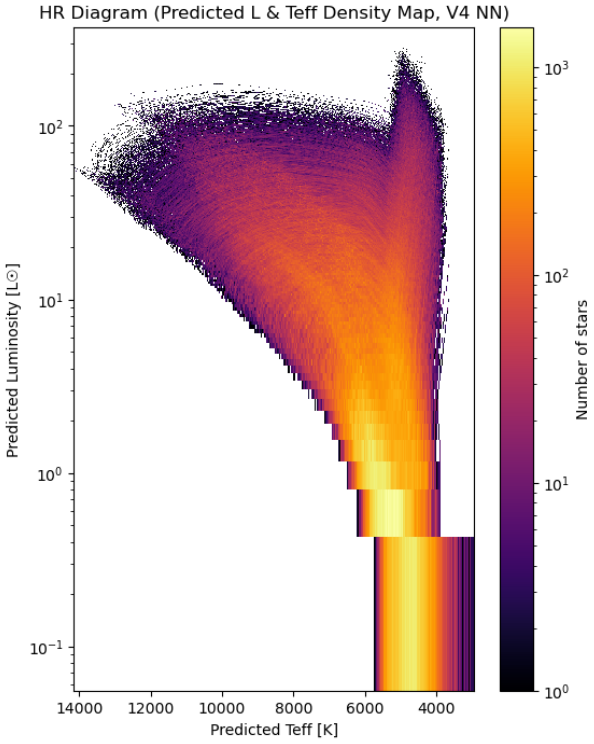


Figure 2: Predicted HR diagram for the NN within the training domain for the simulated stellar tracks dataset. Both models reproduce the expected stellar evolution trends, and results are consistent across both datasets.

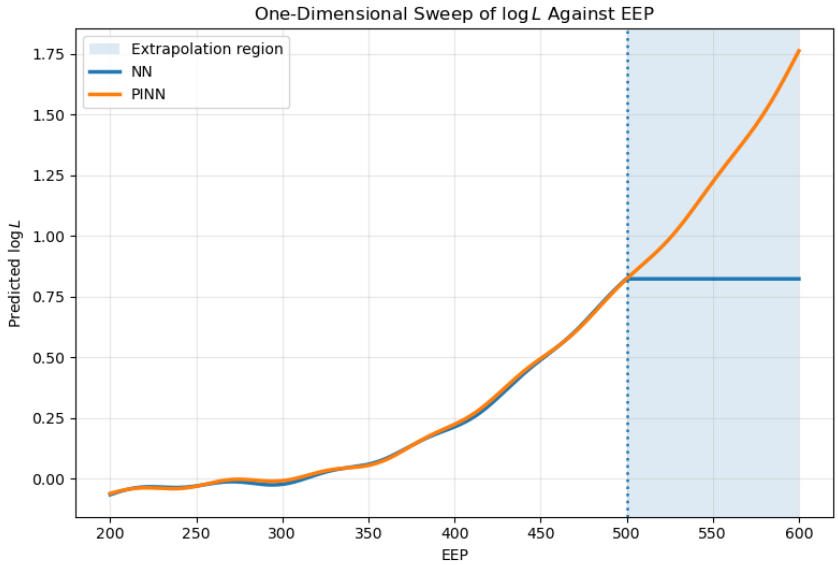


Figure 3: One-dimensional input sweep showing predicted log L as a function of EEP, with all other input variables held constant. The shaded region indicates the extrapolation regime.

Figure 2 shows the predicted HR diagram for the standard NN. Both models produce similar structures, which are consistent with expected stellar evolution behaviour. Figure 3 shows a representative one-dimensional input sweep, where the predicted luminosity is plotted against EEP, with all other input parameters held fixed. Within the training domain, both models produce smooth and continuous predictions, indicating physically consistent behaviour. In contrast, differences become apparent outside the training domain. In both the interpolation and extrapolation regimes, the standard NN produces simplified predictions, with the output tending towards an approximately linear relationship in regions where data is not present, resulting in less physically consistent behaviour. The PINN, however, maintains smooth, nonlinear variation across the input range, producing predictions that remain consistent with expected physical behaviour.

5 Discussion

5.1 Overview of Findings

The results demonstrate a clear distinction between the predictive behaviour of the standard NN and the PINN, highlighting a trade-off between numerical accuracy and physical consistency. The standard NN achieves the lowest error within the training domain, reflecting its ability to flexibly fit the data, given its purely data-driven nature. However, this flexibility becomes a limitation when generalising beyond the training domain. In both the interpolation and extrapolation regimes, the standard NN produces simplified predictions that tend towards approximately linear relationships in regions where training data is not present, indicating limited generalisation beyond the observed data. In contrast, the PINN, while exhibiting slightly higher prediction error within the training domain, beats the standard NN in predictive accuracy in the interpolation and extrapolation achieves higher predictive accuracy than the standard NN in the interpolation and

extrapolation regimes, while maintaining smooth, physically consistent relationships in its predictions. This discrepancy is more pronounced in the extrapolation regime.

In addition to this improvement in generalisation ability, the physics-informed model produces systematically smoother predictions, as reflected by reduced first- and second-difference metrics, indicating lower local variability and more consistent curvature. This behaviour is consistent with the expectation that stellar properties vary continuously with respect to their governing parameters and therefore represents an important indicator of physical consistency.

Visual diagnostics further support these findings: predicted-versus-true plots confirm the differences in prediction error between the models, while HR diagram comparisons and one-dimensional prediction sweeps demonstrate that the physics-informed model produces more physically consistent and smoothly varying outputs. Taken together, these results indicate that while purely data-driven models are effective for fitting known data, the incorporation of physics-inspired constraints provides a clear advantage in terms of and physical plausibility, particularly when generalising to unseen regions of the parameter space.

5.2 Predictive Performance and Generalisation

The results show a clear difference in model behaviour across the different data regimes. While the standard NN achieves the lowest prediction error within the training domain, the PINN provides improved predictive accuracy in both the interpolation and extrapolation regimes. This indicates that the two models differ not only in their ability to fit the training data, but in how they generalise beyond it. This behaviour can be understood from the optimisation strategy of the standard NN. The model is trained purely to minimise the discrepancy between predicted and true values, without any constraints on the form of the function it learns, allowing it to remain highly flexible and closely fit the relationships present in the data. As a result, within the training domain, this flexibility enables the model to accurately capture complex relationships and achieve low prediction error. However, when predictions are required in regions where training data is not present, the model no longer has direct data guidance and must rely on how the learned function extends beyond the observed region. In the absence of any physical constraints, this extension is governed by the behaviour learned near the boundary of the training data and approximates linear behaviour, resulting in constant predictions that are not physically consistent. As a result, the standard NN performs well when data is available but exhibits limited generalisation in regions where data is sparse or absent.

In contrast, the behaviour of the PINN arises from the additional constraints imposed during training. Unlike the standard NN, the PINN is optimised using a composite loss function that includes both the data loss and a physics-inspired smoothness term, which restricts the range of functions the model can represent. As a result, when predictions are required outside the training domain, the model does not extend the learned function in an unconstrained manner. Instead, the smoothness constraint guides the extension of the function, preventing the linear behaviour observed in the standard NN. This leads to predictions that remain consistent with expected physical behaviour and improves predictive accuracy in both interpolation and extrapolation regimes. However, the inclusion of the physics-based constraint also introduces a trade-off during training. Even within the training domain, the model is not optimised solely to minimise the data-fitting loss,

but must also satisfy the smoothness constraint. As a result, the model may not fit the training data as closely as a purely data-driven NN, leading to slightly higher prediction error within the training domain.

5.3 Physical Consistency of Predictions

While the previous section has focused on predictive performance and generalisation, it is important to recognise that minimising prediction error alone is not sufficient in a physical modelling context. In many scientific applications, including stellar modelling, the validity of a model depends not only on its numerical accuracy, but on whether it produces behaviour that is consistent with known physical principles. In this work, physical consistency is assessed through the smoothness of the predicted relationships and their agreement with established stellar behaviour. While smoothness alone does not guarantee that a model is physically correct, it provides a strong indication of physical plausibility in systems where the underlying processes are expected to vary continuously. The physical consistency of the model predictions is first assessed using HR diagrams, which provide a direct comparison between the predicted stellar structure and the expected behaviour from stellar evolution theory. Both the standard NN and the PINN reproduce the overall structure of the HR diagram, indicating that the predicted relationships are broadly consistent with expected stellar evolution trends and providing confidence that the models are capturing the underlying physical behaviour correctly.

To assess this more rigorously, smoothness metrics are used to quantify how the model predictions vary across the input space. The results show that the PINN produces lower first- and second-difference values compared to the standard NN, indicating smoother predictions with more consistent curvature. This is physically significant, as stellar properties are expected to evolve smoothly with respect to their governing parameters. Models that exhibit large variations in curvature may introduce non-physical behaviour, even if their overall prediction error is low. The reduced second-difference values observed in the PINN therefore indicate that it produces predictions that are more consistent with the expected continuous nature of stellar evolution, providing stronger evidence of physical consistency than the standard NN.

This behaviour is further supported by one-dimensional input sweeps, which provide a local view of how the model predictions vary with respect to individual input parameters. The PINN produces smooth and continuous trends across the input range, consistent with the expected behaviour of stellar evolution. In contrast, the standard NN exhibits less consistent behaviour, particularly in regions of sparsely sampled data, giving unphysical constant predictions. This highlights that the incorporation of physics-inspired constraints enables the PINN to generalise more effectively beyond the training data while maintaining behaviour that is consistent with the expected physical properties of the system.

5.4 Limitations

A key limitation of this study lies in the simplified representation of physical constraints within the PINN. Rather than incorporating the full set of governing stellar structure equations, which consist of coupled differential equations describing energy transport, hydrostatic equilibrium, and nuclear processes, the model instead applies a smoothness-

based constraint. While, in principle, the use of full governing equations would provide a more rigorous physical constraint, such equations are highly complex, computationally intensive, and not readily incorporated into a tractable NN training framework for this problem.

A further limitation of this study arises from the use of simulated stellar evolution data rather than real observational data. While the datasets provides a comprehensive and physically consistent representation of stellar behaviour, it is generated from theoretical models and therefore does not fully capture the uncertainties and imperfections present in real observations. In particular, observational data are often affected by measurement noise, incomplete sampling, and systematic biases, none of which are explicitly represented in the simulated datasets. As a result, the learning task considered here is inherently more structured and less noisy than in practical astrophysical applications. This may lead to an optimistic assessment of model performance, particularly in terms of predictive accuracy and smoothness, and therefore the extent to which these results translate to real-world observational settings remains uncertain.

Moreover, a further limitation relates to the optimisation and training procedure of the models, particularly for the PINN. Training PINNs is inherently more challenging than training standard NNs due to the need to balance multiple competing loss terms, including the data loss and the physics-based constraint. In this study, achieving stable and effective training required careful selection of hyperparameters, including the relative weighting of the physics term, the learning rate, and the training schedule. The use of a scheduled weighting scheme was necessary to ensure convergence, allowing the model to first learn from the data before progressively enforcing the physics constraint. In addition, the inclusion of derivative-based constraints significantly increases computational cost, resulting in substantially longer training times compared to the standard NN. This limits the extent to which extensive hyperparameter tuning and repeated experiments can be performed, and therefore the reported results reflect a constrained exploration of the possible model configurations. As a result, the optimisation process remains sensitive to these choices, and different configurations may lead to variations in performance.

6 Conclusion

This dissertation investigated the use of NNs for modelling stellar properties, with a particular focus on comparing a standard NN with a physics-informed variant. The primary aim was to evaluate the extent to which incorporating physics-based constraints improves model performance, not only in terms of predictive accuracy but also in terms of generalisation and physical consistency. Using simulated stellar evolution data, both models were trained under equivalent conditions and evaluated across training, interpolation, and extrapolation regimes, enabling a systematic comparison of their ability to reproduce known data and generalise to unseen regions of the parameter space. The results demonstrate a clear distinction between the two modelling approaches. The standard NN achieves the highest predictive accuracy within the training domain, reflecting its flexibility in fitting the observed data. However, its performance degrades significantly in interpolation and, more notably, in extrapolation, where the absence of data leads to unreliable predictions. In contrast, the PINN exhibits slightly higher error within the training domain but maintains more stable and accurate performance across unseen regions of the parameter space. In addition, the physics-informed model produces smoother

predictions with more consistent curvature, indicating behaviour that is more aligned with expected physical trends. Overall, these findings demonstrate that model performance in a physical modelling context cannot be evaluated solely on the basis of predictive accuracy. In astrophysical applications, where data are often sparse, incomplete, or limited to specific regions of the parameter space, the ability of a model to produce physically consistent behaviour is essential. While purely data-driven NNs are highly effective at fitting known data, they may fail to generate reliable predictions when extrapolating beyond the training domain. In contrast, the results of this study show that the incorporation of physics-informed constraints provides a clear and consistent improvement in model behaviour, particularly in terms of generalisation. Although this comes with a small reduction in training accuracy, the PINN produces predictions that are more stable and more closely aligned with expected physical behaviour. This demonstrates that, for scientific modelling tasks, physics-informed approaches offer a more reliable and physically meaningful alternative to purely data-driven NNs. Future work could extend this study by incorporating more advanced forms of physical constraints within the NN framework. While directly embedding the full set of governing stellar structure equations remains challenging due to their complexity and computational cost, exploring partial or approximate formulations may provide a more complete representation of the underlying physics than the simplified smoothness-based approach used here. In addition, applying the models to real observational data would allow their performance to be evaluated under more realistic conditions, including the presence of noise, measurement uncertainty, and incomplete sampling. Finally, the use of high-performance computing resources, such as the University of Birmingham's BlueBEAR supercomputer, would enable more extensive training and systematic experimentation, allowing for more thorough optimisation of hyperparameters, exploration of alternative architectures, and investigation of training strategies to better balance predictive accuracy with physical consistency.

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